



Full length article

How important are realistic building lifespan assumptions for material stock and demolition waste accounts?

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ARTICLE INFO

Article history:

Received 29 July 2016

Received in revised form 19 January 2017

Accepted 24 January 2017

Available online 21 February 2017

Keywords:

Modelling

Lifespan analysis

MFA

Material stock analysis

Distribution

Buildings

Demolition

ABSTRACT

Accurate assessments of construction materials stocked in the built environment have received increased attention in the Industrial Ecology literature over the past few years. Many recent models that estimate building material inflows, stock accumulation and end-of-life waste, however, rely on simplistic assumptions about the lifespan of built infrastructure. While several probability distributions have been proposed (normal, Weibull, log-normal, and so on) there is no agreement on which model is best suited for modelling the accumulation of building material stock at urban and national levels. In this study we introduce an analysis of the hazard rate of buildings and discuss alternative distribution functions to model lifespan, testing the fit of five commonly used distributions to real data from the cities of Nagoya (Japan), Wakayama (Japan), and Salford (UK). The results highlight how cities with fast replacement rates are overall best modelled by right-skewed distributions, but single cohort levels express independent behaviours based on their characteristics. We investigate the sensitivity of a top-down stock accumulation model to the choice of different distributions and input parameters uncertainties. The results show that different lifespan distribution functions result in very similar overall stock accumulation at the national level, but have large impacts on calculated demolition waste flows. Differences are more pronounced for cities and the choice of a certain distribution will significantly affect the calculation of the average lifetime. Our results suggest that top-down national material stock accounts have high reliability, and are only weakly affected by the choice of one distribution over another. For cities, it is beneficial to use a distribution based on the characteristics of the buildings analysed, with regard to density and building characteristics. Stock accumulation research would profit from future bottom-up research into building lifespans to validate top-down estimation procedures.

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1. Introduction

Studies of material throughput have become a common feature of the Industrial Ecology literature over the past two decades (Adriaanse et al., 1997; Matthews et al., 2000; Fischer-Kowalski et al., 2011). Some materials, such as biomass or fossil fuels, are usually quickly consumed, and rapidly leave the economic sphere in the form of waste and emissions, while others stay within the economy for much longer periods. Of those, construction minerals are probably the materials which have the longest use phase and lifespan. A recent study by Haas and colleagues showed that over 99% of the 24 Gt of construction materials that entered the economy

in 2005 ended as material stock (Haas et al., 2015). In recent years an increasing number of studies have attempted to calculate stocks of several key materials at different levels adopting different methodologies (for an extensive review see Müller et al., 2014; Tanikawa et al., 2015). Accounting for accumulated material stock through this approach is apparently trivial: inflows + domestic extraction – outflows + a number of correction factors¹ = net addition to stock. It is then sufficient to repeat this “simple” equation for a sufficiently long time series to calculate total (in-use) material stock. Nevertheless, despite its apparent simplicity, solving this formula is far from trivial. Robust and reliable data for outflows from the con-

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struction sector, i.e. construction demolition waste data, is hard to come by (Thomsen and van der Flier, 2009). In rare cases, demolition data is collected at the national level, allowing for analysis of trends and the spatial distribution of demolitions for the whole country (Huuhka and Lahdensivu, 2016). In other cases, researchers have managed to extrapolate detailed data of construction and demolition activities at the city level (Aksözen et al., 2016b). A possible strategy to overcome data limitation is creating models which simulate waste outflows, but this obviously poses new challenges in terms of data requirements, and modelling assumptions and limitations. It is generally agreed that a better understanding of the dynamics that characterise the construction and demolition of buildings is a necessary step towards a more sustainable built environment (e.g. Aksözen et al., 2016a; Allen and Hinks, 1996; Thomsen and van der Flier, 2009; Thuvander et al., 2015).

Tanikawa and colleagues identified four different approaches to material stock accounting, including bottom-up accounts, top-down accounts, demand-driven accounts, and accounts based on remote sensing technologies (Tanikawa et al., 2015). Each of these methodologies has its own strengths and weaknesses. Bottom-up accounting relies on item inventories and materials intensities (e.g. Wiedenhofer et al., 2015). At times this information is supported by spatial maps, making the stock geographically explicit, but involves a very time-consuming compilation process (e.g. Lichtensteiger and Baccini, 2008; Tanikawa et al., 2015; Tanikawa and Hashimoto, 2009). The top-down method looks at the stocks using inflow statistical data and imposing a lifetime distribution on the depreciation of stock. While this method is relatively quick to compile it loses information on the spatial distribution of the stock and relies on the goodness of the chosen lifetime distribution and parameters (e.g. Fishman et al., 2014; Hashimoto et al., 2007; Hatayama et al., 2010). Demand-driven accounts use a series of parameters such as population, average household size and material intensity to derive inflows, stock, and outflows of materials (e.g. Müller, 2006; Pauliuk et al., 2013; Vásquez et al., 2016). Remote sensing accounting uses satellite images to investigate stock levels and, despite not reaching the popularity of other methods – mainly due to the lack of validation of the accounts against independent data, could be the only viable option for countries where material flow statistics and detailed maps are not available (e.g. Liang et al., 2014; Rauch, 2009; Yoshida et al., 2016). Among these four methodologies, two depend on lifespan assumptions: the top-down approach, and the demand-driven approach.

Most stock accounts available in the scholarly literature have used top-down and demand driven approaches, and relied on assumptions regarding lifespan distribution to model demolition outflows in some way. Müller and colleagues identified that dynamic material flow analysis used to account for metal flows and stocks mainly relied on the Dirac delta distribution and the Weibull distribution (Müller et al., 2014). Accounts of construction material stocks relied mainly on the normal distribution (e.g. Fishman et al., 2014; Müller, 2006) and Weibull distribution (e.g. Cai et al., 2015; Sandberg et al., 2014).

The main issue, common to all the approaches which rely on lifespan data, is that, up to today, a proper cohort-based dataset is yet to be available, especially for non-residential buildings and infrastructure. A comprehensive comparison of the effect of different distribution functions on stock accounts is yet to be done and is the main aim of this research. We ask:

- How does the choice of a specific lifespan distribution affect the stock account?
- How does this choice influence the forecast of demolition waste?
- Can we identify a demolition probability distribution function that better fits the lifetime of buildings in the real world?

To answer these questions, we first present a theoretical discussion on modelling lifespan distribution, starting from the concept of hazard rate and how this applies to buildings. We then discuss the limitations of some popular probability distribution functions that have commonly been used in lifespan modelling. We demonstrate how the choice of different lifespan distributions and parameters affects the material stock account and waste flow account in a case study of building stock in Japan and the United States. Finally, we test the sensitivity of our results by changing the lifespan distribution parameters to identify those parameters that most affect the results.

2. Calculating building lifespan of real world data

A widespread interest in lifespan analysis arose during the 1970s, when several fields, including engineering, electronic component manufacturing, medicine, and insurance, gained interest in the systematic and scientific analysis of the mortality of particular artefacts, products and people. From the 1980s specific software, such as SAS (SAS Institute) and S-Plus (Statistical Sciences Inc., now TIBCO Software Inc.), became available to study lifetime data. The research field of survival analysis emerged quickly after that.

With lifespan, or lifetime, in its broadest sense, we indicate the period that elapses between two events: from birth to death of a living being, from the production of an engine till its failure, from the construction of a building till its demolition, or even from university graduation till first employment. While common sense suggests that lifespan is something that measures cradle-to-grave intervals, its meaning in the context of survival analysis is much broader.

The lifespan of a single building can be calculated as the difference between the year of its construction and the year of its demolition, whether deliberate or caused by a natural disaster. For objects such as buildings and infrastructure the number of years has been chosen as a convenient unit of time, on the one hand because it is often impossible to retrieve more detailed information (month and day of completion and demolition are usually non-reported, or not disclosed), and on the other because these types of objects have lifespans in the order of decades – sometime centuries – and accounts in months or days would be redundant.

This section will not explain in detail all the methods available to model lifespan. Readers interested in survival analysis will find plenty of literature available (e.g. Lawless, 2011), but we will limit it to the explanation of the concept of hazard rate. This will be followed by the analysis of lifespan distribution models and relative hazard rate, as widely used and suggested in the literature.

2.1. Hazard rate

The hazard rate, or hazard function, is a mathematical expression that indicates the instantaneous rate of death of a unit of interest at a given time t ; formally it is defined as:

$$h(t) = \lambda = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T \leq t + \Delta t | T \geq t)}{\Delta t} \quad (1)$$

Reminding that $\Pr(A|B)$ is the probability of A under the condition B, the previous equation indicates the probability of the event T happening in the interval between t and $t + \Delta t$, given that T did not happen before t , divided by Δt . The normalisation by Δt is important since it makes two hazard rates comparable even if they have been calculated using two different time intervals. It is important to point out that the hazard rate function is not a probability distribution, and that its unit of measure is time^{-1} .

An alternative way of expressing the hazard rate is:

$$h(t) = \lambda = \frac{f(t)}{S(t)} = \frac{f(t)}{1 - F(t)} \quad (2)$$

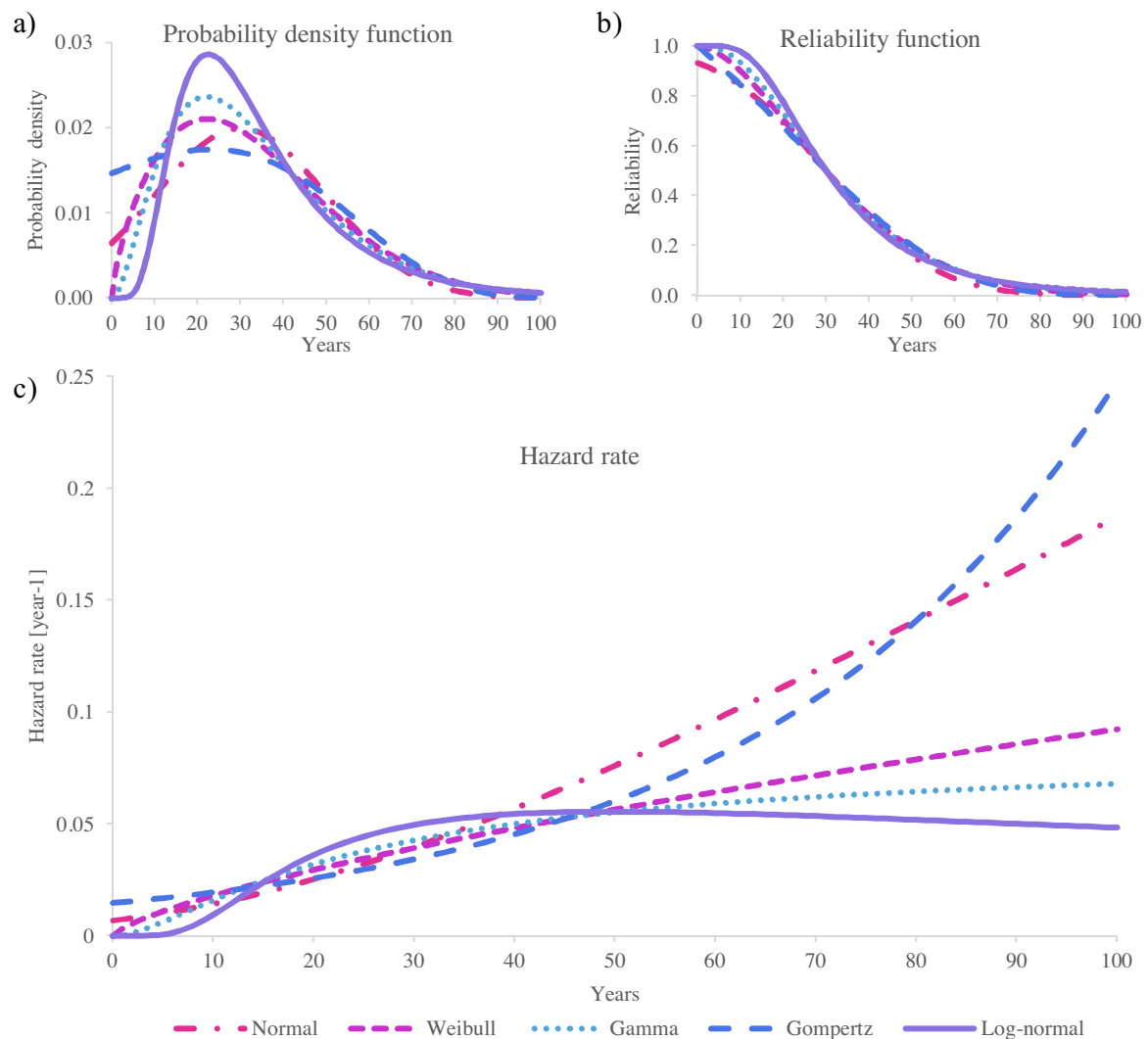


Fig. 1. a) Probability density function for five different distributions. b) Reliability function for five different distributions. c) Hazard rate for five different distributions. All the functions have been modelled to have median = 30 years, and standard deviation = 20 years.

where $f(t)$ is the probability density function (PDF), $S(t)$ is the reliability function (RF), and $F(t)$ is the cumulative density function (CDF).

In simple terms, the hazard rate is a function that expresses the likelihood of failure occurring at a certain time, expressed as *failures per unit of time*. The proper unit of time, seconds, days, years, etc., depends on the specific application for which the function is applied.

2.2. Commonly used probability distributions in material stock analysis

In the past few years several studies have attempted to calculate material stocks by employing a top-down accounting approach² and using different distributions. Some researchers opted for a normal distribution (Fishman et al., 2014; Vásquez et al., 2016); others chose a right-skewed distribution: Weibull (e.g. Bradley and Kohler, 2007; Cai et al., 2015), Log-normal (e.g. Hashimoto et al., 2007; Tanikawa et al., 2015), or Gamma (e.g. Kapur et al., 2008); while others applied a left-skewed Gompertz distribution (e.g. Sarkar et al.,

2011). For a brief description of distribution skewness refer to the Supporting Information (SI 1.4). Among these commonly used distributions, the one which best fits stock accumulation modelling is still unclear. Opting for one over another might seem a small decision at a superficial glance, but our results display wide variability, with considerable effects for the speed of stock accumulation, and the forecast of waste production.

Fig. 1a demonstrates how the five distributions discussed above are shaped when modelling them with the same median (30 years) and standard deviation (20 years). While the Weibull, Gamma, and Log-normal distributions show a PDF of 0 at the year 0, the Normal and Gompertz distributions shows a positive value from the very beginning. This means that, with these particular parameters, the total area below the PDF of these two equations is below 1, assuming therefore that a part of the stock is removed right after construction. This is a modelling issue particularly relevant in the case of short-lived buildings, when modellers have to decide whether to compress the distribution towards the mean in order to avoid cutting part of the left tail, or vice versa to simulate a more spread demolition over time at the cost of assuming that part of the stock is removed straight after its construction.

Fig. 1b illustrates the reliability function for the previously shown five distributions applying the same parameters. Modelled

² For a detailed explanation of this and other stock accounting methodologies refer to Tanikawa et al., 2015.

with a median of 30 years, they all intersect at the point (30, 0.5). All the curves start at the point (0, 1), except for the normal distribution, which starts at (0, 0.93). In this latter case, based on these assumptions, the normal distribution assumes that roughly 7% of buildings would get demolished within a year of their construction.

Fig. 1c shows the hazard rate for the five distributions, which indicate the instantaneous likelihood of demolition. At year 0, the only functions which do not have a hazard rate of 0 are the normal [$h(t) = 6.9E-03$] and Gompertz [$h(t) = 1.5E-02$] distribution functions. After an initial moment of stabilisation, the normal, Weibull, and gamma hazard rates grow linearly, while the Gompertz hazard rate grows exponentially. This is because the Gompertz distribution is mainly used in modelling the lifespan of living beings, which present an exponentially increasing risk of death. The log-normal distribution is the only function, among these five, that remains flat in the early years, presenting a peak at about 50 years, after which it slowly decreases.

The Supporting Information (SI) contains a detailed description of the parameters of these distributions (SI 1.1), their reliability functions (SI 1.2), and formulas to calculate their mean and standard deviation (SI 1.3).

2.3. Some theoretical considerations on building demolitions

When thinking about the likelihood³ of demolition, common experience suggests that, apart from in cases of catastrophic events or exceptional structural deficiencies, very recently built infrastructure and buildings are very unlikely to get demolished. The peak of demolition likelihood will happen after some time has passed, based on the location, period, typology of the construction, et cetera. Those buildings which survive their demolition peak are most likely to represent the best of their cohort, in terms of both architectural design and structural qualities. These individual items are probably worth preserving, and might even gain a historical bond, crystallising even more their presence in the urban fabric. The reader interested in more theoretical considerations about the building stock is invited to refer to the paper “Understanding obsolescence: a conceptual model for buildings” (Thomsen and van der Flier, 2011).

Given these considerations, the only curve that matches the above description is the log-normal distribution. To verify whether this theoretical assumption holds in the real world, we test the best fitting for these curves using three empirical case studies: Nagoya (Japan), Wakayama (Japan), and Salford (UK).

2.4. Case studies

For brevity, this chapter reports only the results for the totality of buildings of the study areas. Details on the R^2 s, average lifespans, standard deviations, and parameters used to calculate the reliability functions for different building groups (residential, commercial, etc.) and for each single cohort can be found in the Supporting Information (SI 2).

2.4.1. Nagoya (Japan)

Through the 4D-GIS methodology introduced in 2009 (Tanikawa and Hashimoto, 2009), Tanikawa and colleagues were able to establish a database of buildings spanning over a significant number of years. This methodology has been applied to the case study of the Naka-ward of Nagoya, the main commercial and service sector district of the city, which today features more than 70,000 buildings.

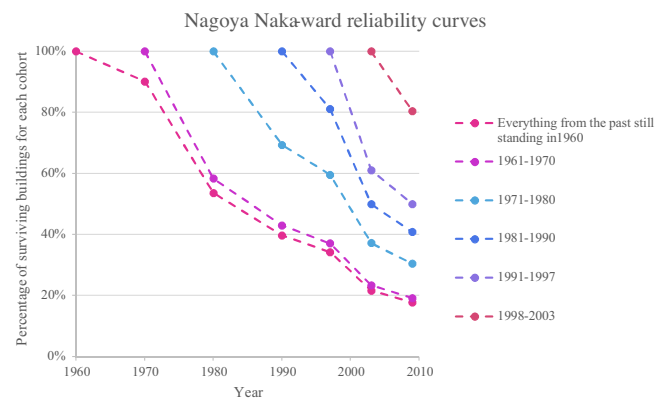


Fig. 2. Survival curves for Naka-ward, Nagoya. Each curve represents a single cohort. The last year of analysis was 2009.

Table 1

Nagoya R^2 values expressed as percentage for curve fitting. The left column represent the year of the cohorts and the sample size, the average R^2 is the mean of all the cohort R^2 for a specific distribution.

Nagoya R^2	Normal	Weibull	Gamma	Gompertz	Log-normal
1960 (n = 28664)	94.5%	98.0%	98.3%	97.1%	98.8%
1970 (n = 29104)	89.3%	98.9%	99.0%	98.3%	98.5%
1980 (n = 26498)	94.8%	98.2%	98.1%	98.4%	97.6%
1990 (n = 26745)	92.2%	97.5%	97.7%	96.6%	98.4%
Average R^2	92.7%	98.1%	98.3%	97.6%	98.4%
R^2 std dev	2.5%	0.6%	0.5%	0.9%	0.5%

The time steps for which building data had been available include 1960, 1970, 1980, 1990, 1997, 2003, and 2009.

The reliability curve for each cohort is shown in Fig. 2, and normalised so that all start from 100%. It is important to point out that each cohort is independent of the others and, for example, the 1970 cohort comprises buildings newly constructed from 1961 to 1970. Using the empirical data, we performed a best fit calculation for the five most commonly used probability distributions, and the results are shown in Table 1.

The log-normal distribution performed the best for the 1960 and 1990 cohorts, while the gamma distribution performed best for the 1970 cohort and the Weibull for the 1980 cohort. All these distributions are right skewed, and match with the theoretical assumption that buildings have an initial demolition-free period, followed by a spike and a long tail. On average the log-normal distribution performed best, both in terms of average R^2 and its relative R^2 standard deviation, while the normal distribution performed the worst for every cohort, even worse than the left-skewed Gompertz distribution. This is probably because the normal distribution, when applied to short-lived buildings, is the only one whose reliability function starts at a value different than 1, id est it assumes that a fairly consistent number of buildings are demolished within their first year after construction, which of course is an unrealistic proposition.

The calculated average lifespans and standard deviations are shown in Table 2.

The log-normal distribution tends to account for shorter lifespan, while the normal distribution assumes – on average – the longest life expectancy. As this case study mainly focused on a commercial area, it shows very quick replacement rates, with much shorter times than the circa 30 years that have been found in other studies focused on Asian countries (Cai et al., 2015; Hu et al., 2010; Tanikawa and Hashimoto, 2009).

There seems to have been an abrupt change in lifespan from 1960 to 1970. While this might have been triggered by a shift in the economic conditions of Japan in its path of industrialisation, it

³ We talk about likelihood, and not probability, because, formally speaking, the hazard rate is not what is mathematically defined as a probability.

Table 2

Nagoya average lifespan and, between parenthesis, standard deviation for four different cohorts and five different distributions. The standard deviation for the Gompertz distribution is reported as (n/a) when this is a complex number.

Nagoya lifespan and std dev	Normal	Weibull	Gamma	Gompertz	Log-normal
1960 (n = 28664)	27.3 (19)	25.2 (21.1)	24.9 (22.3)	25.6 (19.3)	24 (31.2)
1970 (n = 29104)	19.3 (18.5)	14.8 (31.5)	14.5 (29.5)	16.3 (n/a)	14.4 (76.7)
1980 (n = 26498)	14.3 (14.3)	18.6 (19.3)	18.5 (20.7)	18.9 (15.7)	18.1 (37.5)
1990 (n = 23745)	15.2 (9.9)	14.7 (13.3)	14.6 (14.3)	15 (11.4)	14.4 (24.3)

might also be due to the composition of the 1960 cohort. In fact, this cohort includes all buildings standing in 1960, for which the year of construction is from any year until 1960. The 1970 cohort includes all standing buildings that were constructed between 1961 and 1970, de facto greatly limiting the heterogeneity of the sample group.

2.4.2. Wakayama (Japan)

Wakayama is a mid-sized city located in the Kansai region. While the whole municipality has about 370,000 inhabitants, the case study conducted by Tanikawa and Hashimoto focused on the city centre of about 27,000 buildings. The first map that was used for this study dates from 1947, while the last one was from 2004. In total, five snapshots of the city centre were taken in order to obtain stock information for this part of the city. This data was calculated and published in the 4D-GIS paper in 2009 (Tanikawa and Hashimoto, 2009). To calculate the average lifespan, Tanikawa and Hashimoto fitted reliability curves to a logistic curve. We fitted the same data to the five different curves we are analysing, and the results are shown in Table 3.

In the case of Wakayama empirical data, the best overall curve fit is the Weibull distribution, closely followed by the gamma and log-normal distributions. Differences among these curves are in the order of hundredths of percentage point. An order of magnitude above is the Gompertz curve, with an R^2 of 99.0%. The normal and logistic curves represent the data least well, scoring 87.2% and 86.5% respectively. These results confirm what we have previously seen in the case of the Nagoya data, where all the right-skewed distributions perform better than the left-skewed ones, and even better than the symmetric distributions. While the 1947 cohort is best fitted by the Weibull curve, the 1958 cohort is best represented by the log-normal distribution. Unfortunately, the available data did not allow for more cohorts to be studied, diminishing the validity of the results because of a limited number of available vintages.

Table 4 shows the calculated average lifespans using different curves. The right-skewed curves, which are the ones with the lowest errors, present smaller values than the other curves. The highest lifespans are calculated from those curves which present symmetrical tails. There is a 28% difference between the best (log-normal) and worst (normal) fitting curves.

A detailed description on the R^2 , average lifespans, and standard deviation for different building groups (residential, commercial, etc.), and the parameters used to calculate the reliability functions can be found in the Supporting Information (SI 2.3).

2.4.3. Salford, Greater Manchester (UK)

Salford was the second case study described in the 4D-GIS paper (Tanikawa and Hashimoto, 2009). This case, compared to Wakayama, covered a much longer time span, starting in 1849 and ending in 2004, and included nine different time steps. The number of buildings included, however, was smaller than in the other studies with 3240 buildings standing in 2004, which represents 12% of the number of buildings included in the Wakayama sample.

Table 5 displays the results of the curves fitting. Contrary to the results for Nagoya and Wakayama, the distribution that best and most consistently fits the data is the Gompertz distribution,

Table 3

Wakayama R^2 values expressed as percentage for curve fitting. The left column represent the year of the cohorts and the sample size, the average R^2 is the mean of all the cohort R^2 for a specific distribution. The logistic curve R^2 are derived from Tanikawa and Hashimoto, 2009.

Wakayama R^2	Normal	Weibull	Gamma	Gompertz	Log-normal	Logistic
1947 (n = 12428)	88.5%	99.9%	99.9%	98.9%	99.9%	88.3%
1958 (n = 16842)	85.9%	99.9%	99.9%	99.1%	99.9%	84.6%
Average R^2	87.2%	99.9%	99.9%	99.0%	99.9%	86.5%
R^2 std dev	1.9%	0.04%	0.04%	0.2%	0.04%	2.4%

followed by the normal, logistic, and Weibull distributions, while gamma and log-normal scored last. Despite this, the absolute difference between the best and worst fitting distributions is very small, at 1.231%. The reason why the Gompertz distribution best fit the data might be found in the way the City of Salford developed. After a quick development phase thanks to a local coal mine, the city entered a period of stagnation due to the great economic crisis of the 1930s, when many people lived on the verge of poverty in apartments that would nowadays be considered slums. By 1933, a decision to demolish and sanitise the area was taken. What occurred in Salford is hence not natural demolition due to aging, but a deliberate external intervention that terminated the life of many buildings prematurely, a historical specificity which is better modelled by a left-skewed distribution.

The calculated lifespan shows relatively consistent results across the five different distributions, with the exception of the values reported in the 2009 paper by Tanikawa and Hashimoto, which shows values that are on average 20% higher than the rest of the distributions. The calculations for each single cohort are shown in Table 6.

These findings suggest that, in general, right skewed distributions, especially the log-normal, represent the dynamics of stock accumulation and demolition in cities well but does not cater so much for unexpected events such as natural hazards, climate impacts and planning decisions that shorten building lifespans. Do these findings at the city scale hold when we move to the national scale? We aim to address this question by using long-term national stock accumulation data for Japan and the United States.

3. Sensitivity analysis of different distribution curves and input parameters to a material stock accumulation and forecast demolition waste model

A previous study (Fishman et al., 2014) introduced a novel methodology to estimate the accumulated material stock of a nation based on data for material inflows and assumptions about lifetime distribution (top-down inflow driven stock accumulation model). The model uses construction material inflows as input; median, standard deviation, and a normal distribution as exogenous variables; material stock and demolition waste are the outputs. In the following sections we will perform a sensitivity analysis on the model to assess its response to random inputs and variables.

Table 4
Wakayama average lifespan and, between parenthesis, standard deviation for two different cohorts and six different distributions. The standard deviation for the Gompertz distribution is reported as (n/a) when this is a complex number. The logistic curves values are taken from Tanikawa and Hashimoto 2009.

Wakayama lifespan and standard dev	Normal	Weibull	Gamma	Gompertz	Log-normal	Logistic
1947 (n = 12428)	26.7 (25.7)	20.5 (46.6)	20.6 (42.4)	22.5 (n/a)	19.6 (122.6)	23.0 (28.2)
1958 (n = 16842)	28.7 (25.2)	24.3 (54.9)	24.4 (49.4)	25.5 (n/a)	23.9 (154.2)	27.0 (28)

Table 5
Salford R^2 values expressed as percentage for curve fitting. The left column represents the year of the cohorts and the sample size, the average R^2 is the mean of all the cohort R^2 for a specific distribution. The logistic curve R^2 are derived from Tanikawa and Hashimoto, 2009.

Salford R^2	Normal	Weibull	Gamma	Gompertz	Log-normal	Logistic
1849 (n = 98)	98.0%	97.9%	97.3%	98.3%	96.5%	97.9%
1896 (n = 947)	95.8%	95.7%	95.8%	96.1%	95.8%	95.9%
1908 (n = 220)	88.0%	87.5%	85.9%	89.3%	85.0%	88.8%
1932 (n = 1228)	95.8%	95.1%	94.8%	95.8%	94.2%	95.5%
1953 (n = 509)	97.9%	99.0%	99.4%	97.8%	99.6%	97.1%
Average R^2	95.1%	95.1%	94.6%	95.5%	94.2%	95.0%
R^2 Std dev	4.1%	4.5%	5.2%	3.6%	5.5%	3.6%

3.1. Sensitivity to distribution choice

3.1.1. Material stock accumulation

In the following we test how the model responds to different distributions, modelling them so they have the same median and standard deviation. As inputs we have chosen the inflow material dataset for Japan (Krausmann et al., 2011), which spans from 1878 to 2006, and for the United States (Gierlinger and Krausmann, 2012), which spans from 1870 to 2005. We chose to apply a median of 30 years in the case of Japan, and to double this value in the case of the US, in line with the existing literature about lifespan of construction in Asia and North America (Cai et al., 2015; Hu et al., 2010; Tanikawa and Hashimoto, 2009). The standard deviation has been arbitrarily set at 50% of the average lifespan with the intention of avoiding truncating the normal and Gompertz distributions (cf. Fig. 1a and b).

Fig. 3a depicts the stock accumulation of Japan from 1950 to 2005. The chart shows the stock accumulation from 1950 because the model needs to run for a number of decades to ensure that the unknown stock that was present before the start year of the model – 1878 – would be completely demolished or negligible when our account starts.

Fig. 3b represents the relative differences in stock accumulation assuming the normal distribution as baseline, as chosen by Fishman and colleagues in their research. The Gompertz distribution, accounting for approximately 98% of the 29.0 Gt accounted for through the normal distribution, estimates the total stock of Japan at 28.6 Gt. The log-normal distribution estimates 31.0 Gt, approximately +7% compared to the baseline. In between sit the Gamma distribution (30.4 Gt, +5%) and Weibull distribution (29.7 Gt, +2.5%).

Fig. 3c shows the same estimation procedure for accumulated stock in the United States. Fig. 3d illustrates the relative differences between the five different distributions using the normal as baseline. Despite showing very different patterns of inflow, and relative

stock accumulation, the results for the United States display very similar results to those found for Japan (cf. Fig. 3a). The normal distribution accounts for 111.6 Gt, the log-normal for 119.6 Gt (+7%), the gamma for 117.4 Gt (+5%), the Weibull for 114.6 Gt (+2.6%), and the Gompertz for 110.0 Gt (−1.5%).

3.1.2. Demolition waste outflow

Since the stock accumulation model calculates depreciation of stock, it is possible to estimate the demolition flows that come from the construction sector. The formula to estimate demolition outflows is:

$$Outflow_{y_i} = Stock_{y_{i-1}} + Inflow_{y_i} - Stock_{y_i} \quad (3)$$

Yearly stock values are calculated from the model, while the material inflows are taken from the previously mentioned studies.

Fig. 4a graphs the forecast for demolition waste flows for Japan. While all the flows in 1950 are around 30 Mt/year, they grow 25-fold, reaching approximately 740 Mt/year by 2005. Fig. 4b shows that the different distributions result in a similar level of construction demolition waste in 1950 and 2005 with the 1970s being the time with the largest difference between different models. In the 1970s, the log-normal distribution was 43% below the reference case of normal distribution. The Gompertz distribution was 12% above the base case. Japan experienced a period of very fast construction growth after WWII, a time at which Japan saw its yearly inflows of construction material grow 15-fold (Krausmann et al., 2011). In 1973, the time of the first oil price shock, Japan experienced an abrupt stop to its construction expansion era, resulting in the different distribution functions cohering and producing similar outflow rates.

Fig. 4c and d indicate the results for the case of the US. While Japan shows decreasing inflows from the 1990s onward (Krausmann et al., 2011), allowing the different distributions to cohere and reach similar outflow values, the US presents progressively growing inflows (Gierlinger and Krausmann, 2012), thus keeping the results of the distributions separate. The normal distribution, taken as the reference curve, estimates 1078 Mt for 2005. The log-normal distribution shows the lowest flow value, 931 Mt for 2005, which is 86% of what the normal distribution showed. At the opposite end, the Gompertz distribution estimates 1108 Mt of outflows, 103% of the normal account.

In contrast to the urban examples shown in Section 2.4 Case studies, the top-down stock accumulation model for the two national studies of Japan and the United States presents consistent results for the different distributions. In both cases the Gompertz distribution delivers the lowest stock accumulation result, followed by normal, Weibull, gamma, and log-normal. This order is reversed when looking at outflows. On the stock account side, the choice

Table 6
Salford average lifespan and, between parenthesis, standard deviation for two different cohorts and six different distributions. The logistic curves values are taken from Tanikawa and Hashimoto 2009.

Salford lifespan and std dev	Normal	Weibull	Gamma	Gompertz	Log-normal	Logistic
1849 (n = 98)	92.4 (49.8)	90.3 (48.2)	88.4 (52.6)	93.3 (46.2)	86.5 (61.6)	103 (54.3)
1896 (n = 947)	80.7 (7.4)	80.5 (10.4)	80.6 (7.3)	80.2 (13.7)	80.6 (7.3)	104 (8.8)
1908 (n = 220)	88.4 (37.6)	88.2 (36.6)	88.8 (45.7)	88.2 (31.8)	88.7 (55.3)	94 (39.6)
1932 (n = 1228)	93.9 (40.6)	99.3 (50.2)	103.3 (68)	91.5 (32.5)	108 (117.7)	103 (40.4)
1953 (n = 509)	44.8 (22.2)	44.6 (24.1)	44.3 (27.9)	45.4 (20.4)	44.1 (38.2)	55 (24.2)

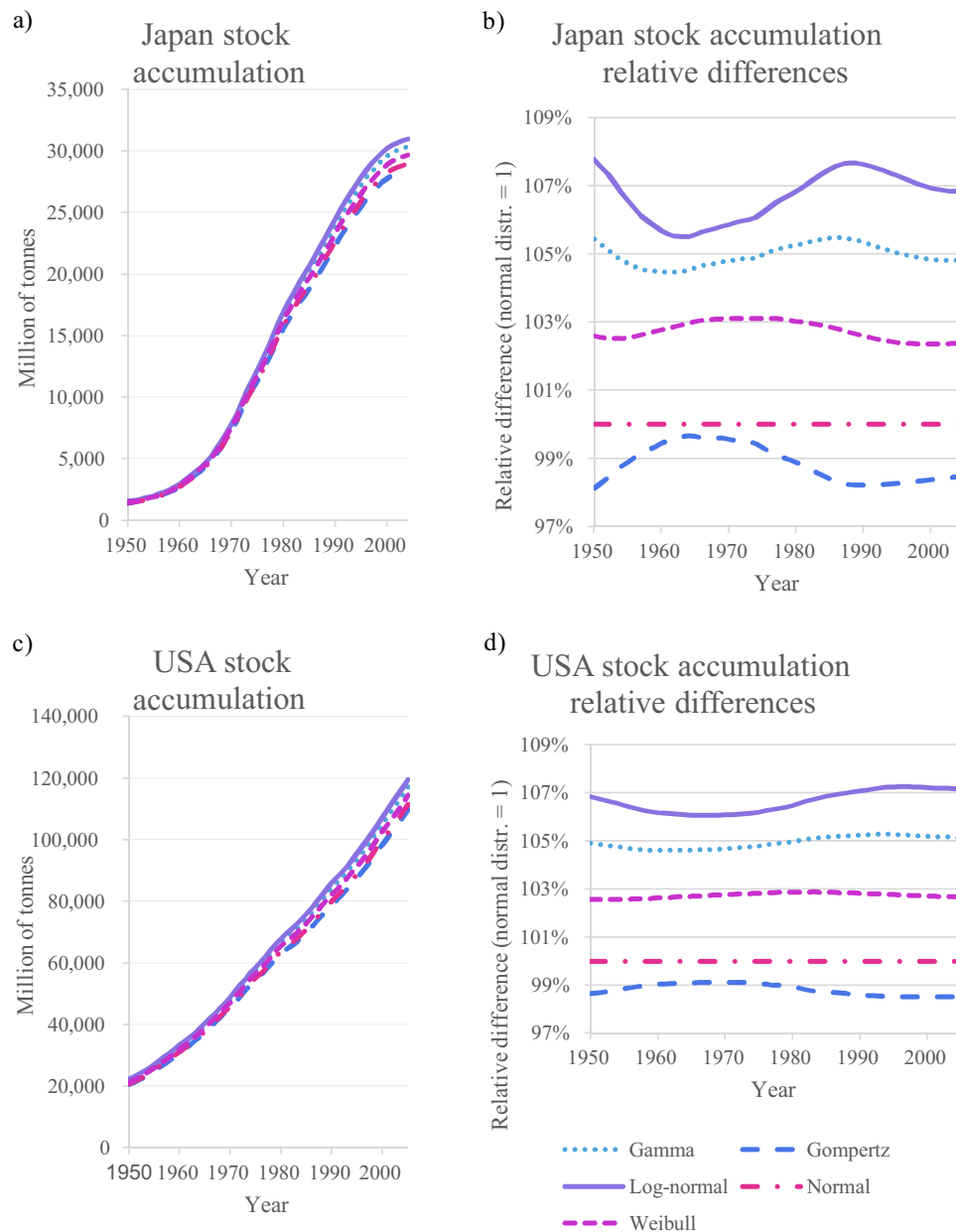


Fig. 3. a) Japan stock accumulation from year 1950–2005. b) Relative differences of the Japan stock accumulation, baseline = normal distribution. c) USA stock accumulation from year 1950–2005. d) Relative differences of the USA stock accumulation, baseline = normal distribution. All the distributions have been modelled to have median = 30 years, and standard deviation = 15 years. Note that the ordinate axis of chart a) and c) have different scales.

of one distribution over another does not affect the results excessively, resulting in a difference of 9% when comparing the highest to the lowest result. When estimating demolition waste flows, however, the choice is much more important, resulting in a difference of 55% between the highest and lowest estimate in the case of Japan.

The United States Environmental Protection Agency reported for the year 2003 a total of 170 Mt of construction and demolition (C&D) materials coming from the building sector (USEPA, 2009). Our model, which estimates not only the building sector but also infrastructure, estimated a flow of $873 \div 1053$ Mt of C&D materials for the same year. This indicates that demolition quantities coming from the infrastructure sector outweigh building demolition materials by a ratio of 4:1. This discrepancy of fivefold is in line with the findings of Kapur and colleagues, who accounted for the year 1996 cement discards from the construction sector equal to $26 \div 29$ Mt,

while, limited to the building sector, the USEPA accounted for 6 Mt of discarded cement for the same year (Kapur et al., 2008).

3.2. Sensitivity of stock estimate to inflows uncertainties

While inflows, being usually calculated a priori, should not be considered a variable of the model, they are often subject to inherent uncertainties (Fischer-Kowalski et al., 2011; Patrício et al., 2015). These uncertainties derive from compilation errors, under-reporting in official statistics, missing data, etc. It was thus decided to extend the sensitivity analysis to the inflows, in order to see how results might change due to errors present in the source data found in statistical yearbooks and databases.

To test the sensitivity of the model to random inflows, we first applied normally distributed random noise to the Japanese material

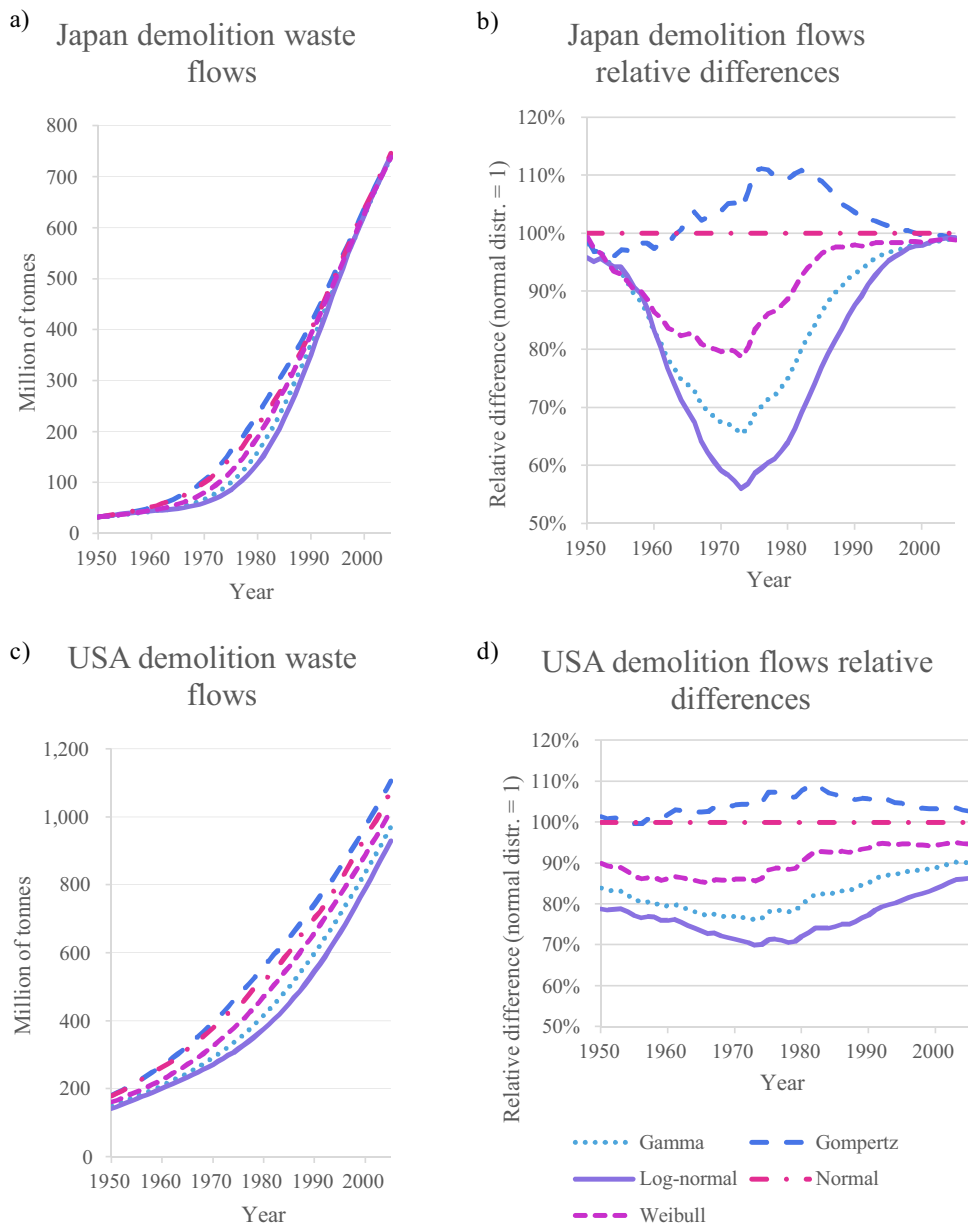


Fig. 4. a) Japan construction waste flows from year 1950–2005. b) Japan relative differences of the demolition waste flows, baseline = normal distribution. c) USA construction waste flows from year 1950–2005. d) USA relative differences of the demolition waste flows, baseline = normal distribution. All the distributions have been modelled to have median = 30 years, and standard deviation = 15 years. Note that the ordinate axis of chart a) and c) have different scales.

inflows, and then performed 10,000 Monte Carlo simulations to assess the stock accumulation results.

The parameters for the Monte Carlo simulations are calculated for each single year, applying a normally distributed uncertainty which has the mean set as the known value, and the standard deviation equal to 30% of that year's value.

The results of the simulation are shown in Fig. 5. The calculated material stock is 31.0 Gt, exactly matching the amount we previously calculated through a deterministic approach (cf. Fig. 3). The coloured bands are calculated by adding and subtracting a single and a double standard deviation respectively. The lowest case accounts for 27.6 Gt, equal to 89% of the average account, while the highest case accounts for 34.5 Gt, equal to 111% of the average, resulting in $\pm 11\%$.

3.3. Sensitivity of stock estimate to the uncertainties of the median parameter of the model distribution function

In order to test how the model responds to random medians, we first applied normally distributed random noise (mean = 30 years, standard deviation = 10 years) to the set median of 30 years, and then performed 10,000 Monte Carlo simulations to assess the stock accumulation results.

The results of this simulation are plotted in Fig. 6. The material stock accounts for 29.7 Gt, missing 1.3 Gt compared to the deterministic account. The bands of confidence show a high degree of uncertainty, ranging from a minimum of 15.1 Gt to a maximum of 44.4 Gt, a difference of $\pm 49\%$.

Stock accumulation sensitivity to inflow uncertainty, case of Japan stock accumulation

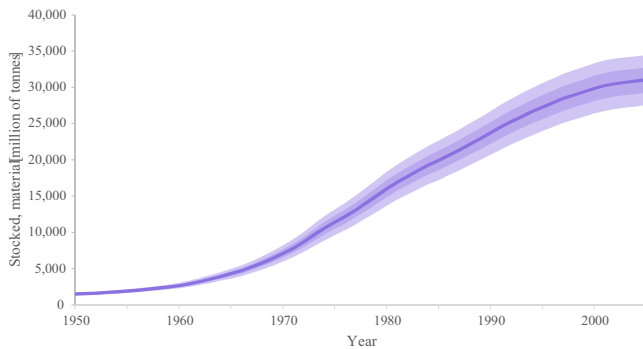


Fig. 5. Construction material stock sensitivity analysis to inflow uncertainties, case of Japan. The main line indicates the average of the Monte Carlo simulation. The coloured bands represent confidence intervals. The model is calculated using a log-normal distribution with median = 30 years, and standard deviation = 15 years. The coloured bandwidths represent one (68%) and two (95%) standard deviations of uncertainty, respectively.

Stock accumulation sensitivity to uncertainties of the median parameter, case of Japan stock accumulation

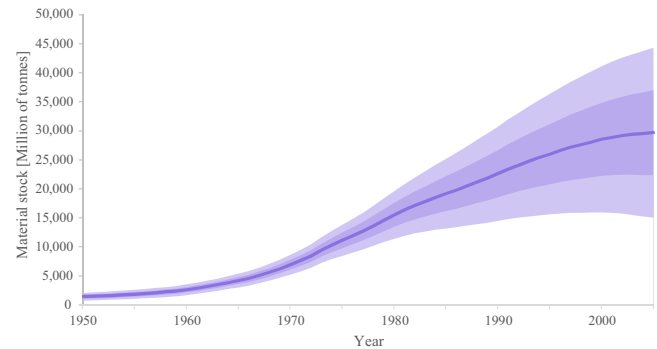


Fig. 6. Stock accumulation sensitivity analysis to uncertainty of the median parameter of the distribution function used in the stock accumulation model. The main line indicates the average of the Monte Carlo simulation. The coloured bands represent confidence intervals. The model is calculated using a log-normal distribution with random median, and standard deviation = 15 years. The coloured bandwidths represent one (68%) and two (95%) standard deviations of uncertainty, respectively.

3.4. Sensitivity of stock estimate to standard deviation parameter of the model distribution function

In order to test how the model responds to standard deviation, we applied normally distributed random noise (mean = 0.434, standard deviation = 0.145) to the scale (σ) parameter of the log-normal distribution. Ideally we would have applied it directly to the standard deviation, but this was not possible without affecting the median. Nonetheless, modifying the scale parameter permits us to alter the standard deviation without affecting the median. We performed 10,000 iterated Monte Carlo simulations to assess the stock accumulation results.

Fig. 7 depicts the results when applying a random standard deviation to the model. The average accounts for 31.0 Gt, matching the deterministic model. The lowest end of the bands of confidence accounts for 30.1 Gt, while the highest reaches 31.8 Gt, resulting in $\pm 3\%$ to the average results.

4. Discussion

4.1. Estimating lifespan of real building data

The curve fitting model showed, unsurprisingly, how the choice of a different distribution can affect the precision of the average lifespan calculation. Short-lived buildings, such as in the case of Nagoya and Wakayama, matched the theoretical assumption that building reliability curves follow a log-normal distribution. The Salford study case showed mixed results: while the Gompertz distribution fits better overall, different building typologies and cohorts are best modelled by different curves (cf. Supporting Information tables S16, S17, S18, S19). It appears that high density residential buildings are best fitted by a log-normal distribution, in the case of industrial buildings the normal distribution shows the lowest results variance and fits the data almost as well as the Gompertz distribution, while the Weibull distribution presents the most consistent results for the tertiary sector. These mixed results might be caused by intrinsically different behaviours of different building typologies when facing longer lifespans, but may also be due to the small numbers of buildings considered in the Salford case study compared to the Japanese studies. Further research in the field of long-lived buildings is necessary to unpack the relation between typology and modelling curve.

Compared to long-lived buildings, short-lived buildings showed greater lifespan estimation differences among different curves.

Plotting demolitions per year on a frequency chart relative to a single building cohort, short-lived buildings show a very long right tail, which matches much better right-skewed distributions (log-normal, gamma, and Weibull). This skewness is much less accentuated in long-lived buildings, smoothing the overall differences between curves, thus making the choice of one distribution over another less significant.

Symmetric distributions showed the poorest fits of all the tested curves, and generally resulted in longer lifespan estimations. While these differences were small in the case of Salford – 80.6 years $\pm 0.8\%$ on average – in the case of short-lived buildings the differences are more pronounced: 18.7 years $\pm 5.0\%$ for the case of Nagoya, and 23.7 years $\pm 9.1\%$ for the case of Wakayama. In the latter case, the difference between the best fitting curve (log-normal) and the poorest one (normal distribution), was equal to 6 years, or 27% of the first account. This could lead to very poor estimations when researchers try to estimate future waste flows from the construction sector, therefore it is advisable to perform a fitting test of several distributions.

In our research, for the Japanese cases, we have been able to calculate lifespan for the past 50–60 years of a fairly wide area, while in the case of Salford we have been able to assess the past 160 years of a limited central area. It is important to remark that recent findings on the whole city of Zurich for the past 170 years by Aksözen and colleagues showed how the lifespan of buildings has decreased over time (Aksözen et al., 2016a), thus researchers interested in modelling long-term stock accumulation should not overlook this aspect.

From a purely mathematical point of view, because of the higher number of data points, the earlier cohorts are drawn more precisely than the latter. The 1960 cohort uses 7 data points to draw the best fitting curve, while the 1990 cohort uses only 4, the minimum amount that we can use to fit a curve.⁴ In the case of functional differences between cohorts, modellers are thus advised to have greater confidence in earlier curves than later ones.

⁴ All the distributions considered in this study are characterised by two parameters, and this means that it is necessary to have a minimum of two points to characterise their parameters. Yet, independently of their parameters, the curves generally start at the point (0,1), which means that a third point it is necessary. Finally, a fourth point is required to have an estimation of the fitting error.

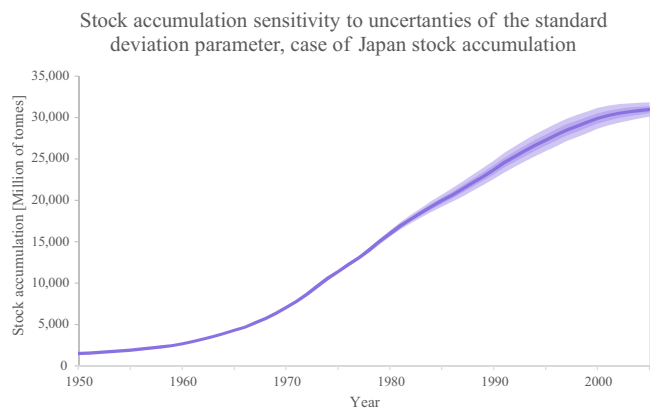


Fig. 7. Stock accumulation sensitivity analysis to uncertainty of the standard deviation parameter of the distribution function used in the stock accumulation model. The main line indicates the average of the Monte Carlo simulation. The coloured bands represent confidence intervals. The model is calculated using a log-normal distribution with median = 30 years, and random standard deviation. The coloured bandwidths represent one (68%) and two (95%) standard deviations of uncertainty, respectively.

4.2. Top-down inflow driven stock accumulation and waste flow forecast model: sensitivity to uncertainties and different lifetime distributions through Monte-Carlo simulation

The top-down inflow driven model was proved to be a viable tool in estimating the total of accumulated construction materials present in a society. The five different distributions showed consistent and relatively close results relative to stock accumulation. They showed, however, very different results when accounting for demolition flows. If policy makers are to plan where to allocate this kind of waste, uncertainty of up to 50% in the sheer quantity of materials needing to be disposed of would lead to terrible outcomes – either greatly underestimating or overestimating total waste amounts.

The reason behind this very different impact of the selection of a specific lifetime distribution function on the stock accumulation and waste flows account is the buffer effect of lifespan. After the end of WWII most developed countries experienced rapidly increasing material inflows (Schaffartzik et al., 2014). These inflows are immediately recorded by the stock accumulation model, and, after an initial period with no depreciation, start to be slowly converted into demolition waste. Lifespan, acting as a buffer between inflows and outflows, delays the transformation of construction materials into waste by a number of years equal to its average. Thus, the outflow of demolition waste recorded today stems from the inflows which entered the system perhaps decades ago. A relative difference of 100 million tonnes between two different lifespan distributions, for instance, represents 0.1% of the US construction material stock in recent years, but almost 10% of the demolition waste.

The model proved to be mainly sensible to median uncertainties ($\pm 49\%$), followed by inflows uncertainties ($\pm 11\%$), and lastly by the standard deviation uncertainties ($\pm 3\%$). Considering the results, modellers should mainly focus on choosing a realistic average lifespan, without worrying too much about the chosen standard deviation. Inflows are inherently subject to account uncertainties, but even so the model responded pretty consistently, proving to be a suitable tool for macro analysis. Standard deviation plays a minor role in the stock accounting model, and any value between 0.75-median and 0.20-median would return realistic results. These results match the findings of Chen and Shi's research about aluminium stocks in China (Chen and Shi, 2012).

Since the model is mainly affected by lifespan (i.e. median parameter), this is where most efforts should be concentrated.

Unfortunately, up to today a large-scale analysis on the lifespan of buildings is still missing, and we can only rely on a small set of case studies which have adopted very different accounting methods. A bottom-up approach, as done in the cases of Nagoya, Wakayama, and Salford, can give very reliable results, and even show how lifespan has evolved over time, but these studies are extremely time consuming in terms of data compilation, and require long computation time to be processed. In some cases researchers have been able to estimate national averages from data in statistical yearbooks (Cai et al., 2015), but only a few nations provide enough data to conduct this type of analysis. Some studies provide yearly demolition rates rather than lifespan (Bradley and Kohler, 2007; Deilmann et al., 2009; Lichtensteiger and Baccini, 2008; Petersdorff et al., 2006; Reyna and Chester, 2015; Thomsen and van der Flier, 2011; Wiedenhofer et al., 2015), but it is not possible to convert yearly demolition rates to an average lifespan without knowing the composition of each cohort, making it impossible to use this information with the top-down inflow driven stock accumulation model. More extensive and detailed research about lifespan is therefore advisable.

4.3. Estimating construction material stocks and related waste flows. Is there a best lifespan distribution?

We found that the log-normal distribution best fitted the cases of Nagoya and Wakayama, matching the theoretical assumption that buildings, when they have passed their peak of hazard, would last for quite a long time. Our results match recent findings on the analysis of lifespan for life cycle inventories (LCI) (Qin and Suh, 2016), showing an interesting correlation between findings in the field of material flow analysis (MFA) and LCI, which could be investigated in future research aiming to link material consumption to environmental impacts. While it might be tempting to immediately jump to the conclusion that the log-normal distribution should be standard, the case of Salford, best fitting a Gompertz distribution, proved that any external factor such as a natural disaster, a sudden economic crisis, or deliberate massive demolitions, invalidate the assumption, and other distribution functions may better fit the reality. Moreover, the lifespan distribution analysis performed in chapter 2.4 Case studies, investigating central areas of three dense cities, showed that in several cases other right skewed distributions performed very well. Further confirmation of our findings requires be supported by future studies assessing wider and more diverse areas (e.g. rural, peripheral, low density, etc).

From this research, we know that the normal distribution does not reflect the reality of building lifespans well, yet we still do not have sufficient data to determine if this distribution could well represent the total stock (building + infrastructure). We also know that the impact of choosing a certain lifetime distribution function on final stock accumulation numbers is negligible, thus researches on construction material stocks conducted by Fishman and colleagues or Vásquez and colleagues, which were performed applying a normal distribution, resulted in verisimilar accounts (Fishman et al., 2014; Vásquez et al., 2016). Had these studies been focused on waste flow accounts as well, their choice of distribution function would have distorted the final result considerably. This demonstrates that both the context of the analysis and the focus on results needs to be considered when choosing a specific lifetime distribution.

Tanikawa and Hashimoto (Tanikawa and Hashimoto, 2009) calculated the average lifespan of buildings for Wakayama and Salford, fitting the data to a logistic distribution, creating results that are on average 15% higher than those estimated by the best fitting distribution for both cities.

These are just some examples where, in light of this study, a more judicious choice of distribution could bring results more

rooted in reality. This would allow them to be used by decision makers with greater confidence, e.g. for planning the disposal of demolition waste, a waste stream that will rapidly grow throughout the twenty-first century (Müller, 2006).

Nevertheless, modellers should be aware that, whereas the log-normal distribution seems to be the most realistic choice out of the five proposed, unpredictable events, such as earthquakes or civil wars, would completely invalidate this assumption. Furthermore, our results come from the analysis of buildings. In order to verify whether this evidence would hold true for infrastructure, future research should aim to assess the lifespan of infrastructure.

5. Conclusions

In this research, we investigated how different lifespan distributions and uncertainties affect the results of a top-down stock accumulation model. Using data from urban case studies we assessed how well different lifespan distributions match real world data. The results showed that the log-normal distribution, and in general right-skewed distributions, works the best for short-lived buildings, which are very typical of Asian cities. The case of Salford showed that its buildings are best matched by a Gompertz distribution, but also gave different results for different building typologies (cfr. SI 2.2), perhaps due to the small size of the study area, or due to the peculiarity of the economical evolution of the specific case study. The results match the theoretical assumption derived from analysis of the hazard rate that the “best individuals” of a single cohort will live much longer than the average.

We then tested how the lifespan model was affected by different distribution choices using national datasets for Japan and the United States and showed how uncertainties in exogenous variables affect results, comparing stock accumulation and demolition waste flow accounts. The results show that different distribution functions lead to very similar figures for stock accumulation, but very different demolition waste flows. Distribution functions that are right skewed, with the majority of demolition happening in the later stage of life, best represent the reality of built environment, indicating a slight preference for log-normal distribution for estimating accumulated stock in buildings.

Applying the most realistic curve becomes vital if researchers are to provide policy makers with robust estimations in order to properly allocate this type of waste. At the same time, modellers should keep in mind that venturing into long-term projection of waste production is a stretch as natural disasters and deliberate demolitions can completely invalidate projections.

Material inflow data have an intrinsic uncertainty, especially in the non-metallic minerals category (cfr. Fischer-Kowalski et al., 2011), and – with the exclusion of the metallic materials category – accounting discrepancies are very common even within the same database. Uncertainty in inflow parameters did not excessively affect the top-down stock accumulation model, proving its viability for macro scale analysis.

We showed that the top-down inflow driven stock accumulation model is mainly sensitive to the average lifespan, and almost insensitive to the choice of lifespan distribution function. This clearly suggests a direction for future research: calculating realistic lifespans for different typologies of buildings. While there is already solid proof that Asian cities have very short lifespans, in the order of 25–30 years, more research on European and American lifespan trends is still needed, and future research should aim to fill this knowledge gap.

Decision makers who are striving to move towards a material sound society should use the analysis of the mean lifespan over time as an indicator to assess and promote policies that encourage reuse, refurbishment, and adaptation of buildings over demolition and

reconstruction. While this latter practice might have short-term benefits to the economy, it would, in the long run, deplete the natural environment of irreplaceable natural resources and increase the emissions of pollutants, resulting in a much worse scenario than a slow construction sector.

Acknowledgements

This research was financially supported by the Environment Research and Technology Development fund (1-1402) of the Ministry of the Environment, Japan. The authors are grateful to Karin Hosking of CSIRO for copy- editing the manuscript and to three anonymous reviewers who, with their thoughtful comments, helped improving the scientific quality of this research.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.resconrec.2017.01.015>.

References

- Adriaanse, A., Bringezu, S., Hammond, A., Moriguchi, Y., Rodenburg, E., Rogich, D., Schütz, H., 1997. *Resource Flows: The Material Basis of Industrial Economies*. World Resources Institute Washington, DC.
- Aksözen, M., Hassler, U., Kohler, N., 2016a. Reconstitution of the dynamics of an urban building stock. *Build. Res. Inf.* 0, 20, <http://dx.doi.org/10.1080/09613218.2016.1152040>.
- Aksözen, M., Hassler, U., Rivallain, M., Kohler, N., 2016b. Mortality analysis of an urban building stock. *Build. Res. Inf.* 0, 1–19, <http://dx.doi.org/10.1080/09613218.2016.1152531>.
- Allen, S., Hinks, J., 1996. How long should housing last?: Some implications of the age and probable life of housing in England. *Constr. Manage. Econ.* 14, 531–535.
- Bradley, E.P., Kohler, N., 2007. Methodology for the survival analysis of urban building stocks. *Build. Res. Inf.* 35, 529–542.
- Cai, W., Wan, L., Jiang, Y., Wang, C., Lin, L., 2015. Short-lived buildings in China: impacts on water energy, and carbon emissions. *Environ. Sci. Technol.* 0, 8, <http://dx.doi.org/10.1021/acs.est.5b02333>.
- Chen, W.-Q., Shi, L., 2012. Analysis of aluminum stocks and flows in mainland China from 1950 to 2009: exploring the dynamics driving the rapid increase in China's aluminum production. *Resour. Conserv. Recycl.* 65, 18–28, <http://dx.doi.org/10.1016/j.resconrec.2012.05.003>.
- Deilmann, C., Effenberger, K.-H., Banse, J., 2009. Housing stock shrinkage: vacancy and demolition trends in Germany. *Build. Res. Inf.* 37, 660–668.
- Fischer-Kowalski, M., Krausmann, F., Giljum, S., Lutter, S., Mayer, A., Bringezu, S., Moriguchi, Y., Schütz, H., Schandl, H., Weisz, H., 2011. Methodology and indicators of economy-wide material flow accounting. *J. Ind. Ecol.* 15, 855–876.
- Fishman, T., Schandl, H., Tanikawa, H., Walker, P., Krausmann, F., 2014. Accounting for the material stock of nations. *J. Ind. Ecol.* 18, 407–420.
- Gierlinger, S., Krausmann, F., 2012. The physical economy of the United States of America. *J. Ind. Ecol.* 16, 365–377.
- Haas, W., Krausmann, F., Wiedenhofer, D., Heinz, M., 2015. How circular is the global economy?: an assessment of material flows, waste production, and recycling in the European union and the world in 2005. *J. Ind. Ecol.* 19, 765–777, <http://dx.doi.org/10.1111/jiec.12244>.
- Hashimoto, S., Tanikawa, H., Moriguchi, Y., 2007. Where will large amounts of materials accumulated within the economy go?—a material flow analysis of construction minerals for Japan. *Waste Manage.* 27, 1725–1738.
- Hatayama, H., Daigo, I., Matsuno, Y., Adachi, Y., 2010. Outlook of the world steel cycle based on the stock and flow dynamics. *Environ. Sci. Technol.* 44, 6457–6463, <http://dx.doi.org/10.1021/es100044n>.
- Hu, M., Bergsdal, H., Voet, E., van der Huppes, G., Müller, D.B., 2010. Dynamics of urban and rural housing stocks in China. *Build. Res. Inf.* 38, 301–317, <http://dx.doi.org/10.1080/09613211003729988>.
- Huuhka, S., Lahdensivu, J., 2016. Statistical and geographical study on demolished buildings. *Build. Res. Inf.* 44, 73–96.
- Kapur, A., Keoleian, G., Kendall, A., Kesler, S.E., 2008. Dynamic modeling of in-use cement stocks in the United States. *J. Ind. Ecol.* 12, 539–556.
- Krausmann, F., Gingrich, S., Nourbakhch-Sabet, R., 2011. The metabolic transition in Japan. *J. Ind. Ecol.* 15, 877–892.
- Lawless, J.F., 2011. *Statistical Models and Methods for Lifetime Data*, 2nd edition. John Wiley & Sons.
- Liang, H., Tanikawa, H., Matsuno, Y., Dong, L., 2014. Modeling in-use steel stock in China's buildings and civil engineering infrastructure using time-series of DMSP/OLS nighttime lights. *Remote Sens.* 6, 4780–4800.
- Lichtensteiger, T., Baccini, P., 2008. Exploration of urban stocks. *J. Environ. Eng. Manage.* 18, 41–48.

- Müller, E., Hilty, L.M., Widmer, R., Schluep, M., Faulstich, M., 2014. Modeling metal stocks and flows: a review of dynamic material flow analysis methods. *Environ. Sci. Technol.* 48, 2102–2113, <http://dx.doi.org/10.1021/es403506a>.
- Müller, D.B., 2006. *Stock dynamics for forecasting material flows—case study for housing in The Netherlands*. *Ecol. Econ.* 59, 142–156.
- Matthews, E., Amann, C., Bringezu, S., Hüttler, W., Ottke, C., Rodenburg, E., Rogich, D., Schandl, H., Van, E., Weisz, H., et al., 2000. *The Weight of Nations—Material Outflows from Industrial Economies*. World Resources Institute, Citeseeer.
- Patrício, J., Kalmykova, Y., Rosado, L., Lisovskaja, V., 2015. Uncertainty in material flow analysis indicators at different spatial levels. *J. Ind. Ecol.* 19, 837–852, <http://dx.doi.org/10.1111/jiec.12336>.
- Pauliuk, S., Milford, R.L., Müller, D.B., Allwood, J.M., 2013. The steel scrap age. *Environ. Sci. Technol.* 47, 3448–3454.
- Petersdorff, C., Boermans, T., Harnisch, J., 2006. Mitigation of CO₂ emissions from the EU-15 building stock beyond the EU directive on the energy performance of buildings. *Environ. Sci. Pollut. Res.* 13, 350–358.
- Qin, Y., Suh, S., 2016. What distribution function do life cycle inventories follow? *Int. J. Life Cycle Assess.*, 1–8, <http://dx.doi.org/10.1007/s11367-016-1224-4>.
- Rauch, J.N., 2009. Global mapping of Al, Cu Fe, and Zn in-use stocks and in-ground resources. *Proc. Natl. Acad. Sci.* 106, 18920–18925, <http://dx.doi.org/10.1073/pnas.0900658106>.
- Reyna, J.L., Chester, M.V., 2015. The growth of urban building stock: unintended lock-in and embedded environmental effects. *J. Ind. Ecol.* 19, 524–537, <http://dx.doi.org/10.1111/jiec.12211>.
- Sandberg, N.H., Sartori, I., Brattebø, H., 2014. Using a dynamic segmented model to examine future renovation activities in the Norwegian dwelling stock. *Energy Build.* 82, 287–295, <http://dx.doi.org/10.1016/j.enbuild.2014.07.005>.
- Sarkar, S., Chamberlain, J.F., Miller, S.A., 2011. A comparison of two methods to conduct material flow analysis on waste tires in a small island developing state. *J. Ind. Ecol.* 15, 300–314, <http://dx.doi.org/10.1111/j.1530-9290.2010.00323.x>.
- Schaffartzik, A., Mayer, A., Gingrich, S., Eisenmenger, N., Loy, C., Krausmann, F., 2014. The global metabolic transition: regional patterns and trends of global material flows, 1950–2010. *Glob. Environ. Change* 26, 87–97.
- Tanikawa, H., Hashimoto, S., 2009. Urban stock over time: spatial material stock analysis using 4d-GIS. *Build. Res. Inf.* 37, 483–502.
- Tanikawa, H., Fishman, T., Okuoka, K., Sugimoto, K., 2015. The weight of society over time and space: a comprehensive account of the construction material stock of Japan, 1945–2010. *J. Ind. Ecol.*, <http://dx.doi.org/10.1111/jiec.12284>.
- Thomsen, A., van der Flier, K., 2009. Replacement or renovation of dwellings: the relevance of a more sustainable approach. *Build. Res. Inf.* 37, 649–659.
- Thomsen, A., van der Flier, K., 2011. Understanding obsolescence: a conceptual model for buildings. *Build. Res. Inf.* 39, 352–362.
- Thuvander, L., Österbring, M., Mangold, M., Mata, E., Wallbaum, H., Johnsson, F., 2015. Spatial exploration of the refurbishment dynamics of urban housing stocks. In: *Proceedings of the Computers in Urban Planning and Urban Management Conference, CUPUM*.
- USEPA, 2009. Estimating 2003 Building-Related Construction and Demolition Materials Amounts (No EPA530-R-09-002). United States Environmental Protection Agency.
- Vásquez, F., Løvik, A.N., Sandberg, N.H., Müller, D.B., 2016. Dynamic type-cohort-time approach for the analysis of energy reductions strategies in the building stock. *Energy Build.* 111, 37–55, <http://dx.doi.org/10.1016/j.enbuild.2015.11.018>.
- Wiedenhofer, D., Steinberger, J.K., Eisenmenger, N., Haas, W., 2015. Maintenance and expansion: modeling material stocks and flows for residential buildings and transportation networks in the EU25. *J. Ind. Ecol.* 19, 538–551.
- Yoshida, K., Fishman, T., Okuoka, K., Tanikawa, H., 2016. Material stock's overburden: automatic spatial detection and estimation of domestic extraction and hidden material flows. *Resour. Conserv. Recycl.*, <http://dx.doi.org/10.1016/j.resconrec.2016.09.010>.

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